Discover the knowledge. Bound the work. Verify the result.

Omega Hive × ASI

Private lookup, bounded cognition and verified chain commitments

THROUGH M5 · LOOKUP + C1–C4 + CHAIN

HIVE · DETTA · PLN · F1R3

Architecture and evidence • 16 October 2026 • supported local scope

ELI16: remember, weigh evidence, do the calculation

Think of a student investigating a question with a notebook, evidence ledger and calculator.

C1 · NOTEBOOK

Keep the original story

Record what happened and where each report came from. Keep contrary observations; replay does not add another event.

C2 · EVIDENCE LEDGER

Know what supports a belief

A copy is still the same source. A conclusion keeps its parents; it does not become a fresh observation.

PLN · CALCULATOR

Apply a guarded formula

Use the existing pinned mathematics on exact inputs. C2 checks and saves the result; the calculator owns no beliefs.

What this gives us

An inspectable trail from original observations to admitted conclusions.

A repeated message, a new conclusion and an independent observation mean different things.

ELI16: a priority list and a limited attempt budget

There are two prepared questions, but time for only one calculation. Which should run first?

C3 · Priority list

Give each eligible task an integer score. Rank it, decay it explicitly, and remember feedback once.

C4 · Work coordinator

Choose from that list. Save the attempt reservation before calculation, then track admission and feedback.

Restart · Same budget

Reopening the notebook does not refill the budget. Recover a saved decision; do not credit it twice.

Observed in real Hive

One native action, one admitted conclusion, one feedback credit; restart and replay tested.

The deliberate tradeoff

A crash can spend the last attempt without producing a conclusion. The receipt says so.

Priority is not truth. A bounded step finishing is not a promise that a conclusion was found.

ELI16: finishing the job and settling its receipts

A notebook entry, a calculation and a reward can finish at different times.

C1 + RELAY

The report was processed

The notebook gets a delivery receipt even when a report cannot change a number. Contrary evidence stays in the story.

C3 · ATTENTION

The reward has one outcome

Ask the priority-list owner: already credited, retired without credit, or still pending. A lost reply never earns a second reward.

C4 + HIVE

The receipt tells both stories

Show the original work result and the later cleanup progress separately. Cleanup does not buy another attempt.

Observed

33 observations drained; 20 new process-kill cases; no new formula calls during reconciliation.

M3.1 settles recorded effects. It does not extend the deadline, change the past or start new thinking.

ELI16: approve a fingerprint, then check the receipt

Imagine signing a sealed envelope containing the fingerprint of one shareable result.

1 · APPROVE

Name the exact result

Hive finishes a generated task. The operator approves that result, its destination and who may sign it.

2 · SAVE, THEN SEND

Keep the same envelope

Save the signed request before sending. After a crash, retry those same bytes; a lost reply does not mean a new request.

3 · VERIFY

Read what was stored

Check the fingerprint in the block’s resulting state and the matching deployment’s finalized effect. Repeat after restart.

Private notebooks stay private

Only the approved generated workflow is eligible. Episodes, beliefs and their identifying hashes stay off-chain.

A stored fingerprint identifies the approved result. It does not establish that the reasoning is true.

ELI16: a catalogue, labelled links and the actual books

Ask the library for matching records, follow their links, then read the stored content.

SEARCH · CATALOGUE

Find matching records

Ask for exact attributes. The catalogue returns references and says how far it has indexed the accepted records.

RELATIONS · LINKS

Follow explicit structure

A record may require three other records. Follow the labelled, ordered roles to discover their references.

ATOMSTORAGE · BOOKS

Retrieve the actual bytes

Read the discovered references through the storage contract. The consumer was never given an answer list.

Observed in real Hive

5 matching records → 15 relations → 8 payloads. Append, catch up, rebuild, restart and replace Search.

“No match in this indexed prefix” and “the index is too far behind” are different answers.

Fifteen modules; four responsibility groups

A module owns a responsibility. It need not be a separate process or repository.

CLIENT / COGNITION

- Episodic memory
- Belief revision
- PLN · one rule
- STI attention
- Inference control

APPLICATION PORTS

- AtomStorage
- Relations
- Search projection
- Agent messaging
- Delegated execution

CONTROL & POLICY

- Registry
- Compatibility resolver
- Authorization

Local test policy
and static descriptors

PROVIDER ADAPTERS

DA storage

Real local DeTTa

Consensus / finality

Real isolated
F1R3node · M4

15 / 15 BOUNDED FAMILIES

IMPLEMENTED LOCAL

REAL PROVIDER

One grounded rule; no ECAN or LTI

Data responsibilities: bytes → references → history

Persistence, identity, structure, discovery and provenance have different owners.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / STATUS
DA storage	Persist opaque bytes put · get · availability	Payloads + shares REAL LOCAL PROVIDER
AtomStorage	Scoped immutable envelopes create · get/getMany · head/readCommits	Reference + replay + accepted feed IMPLEMENTED · M5
Relations	Represent ordered structural roles assert/getRelation · outgoing/incoming	Accepted structures + incidence IMPLEMENTED · M5
Search projection	Find refs at a known watermark query · catchUp · beginRebuild	Index + checkpoint + generation IMPLEMENTED · M5
Episodic memory C1	Record / append / read ordered events pendingEvidence · trusted completion	Private history + delivery intents IMPLEMENTED · M3.1

DA persists bytes. Only the application owners interpret events, claims or beliefs.

Cognitive responsibilities: four separate roles

Attention, belief admission, pure mathematics and work coordination remain separate.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / BOUNDARY
Belief revision C2	submitEvidence · readBelief submitDerivation · readDerivation	Supports, snapshots, attempts Only C2 commits belief changes
PLN kernel	Apply one grounded rule to exact premises profiles · applyRule	No mutable domain state Pure native proposal
Attention C3	select · advance · submitStimulus submitFeedback · resolveFeedback	STI, clock, causal receipts / retirement No cognition dependency
Inference control C4	Run one immutable workset step · readProgress · reconcile	Selection, budgets, closure progress No truth or attention ownership

C4 calls C2 for admission. C2 invokes the kernel, then checks authority and commits its decision.

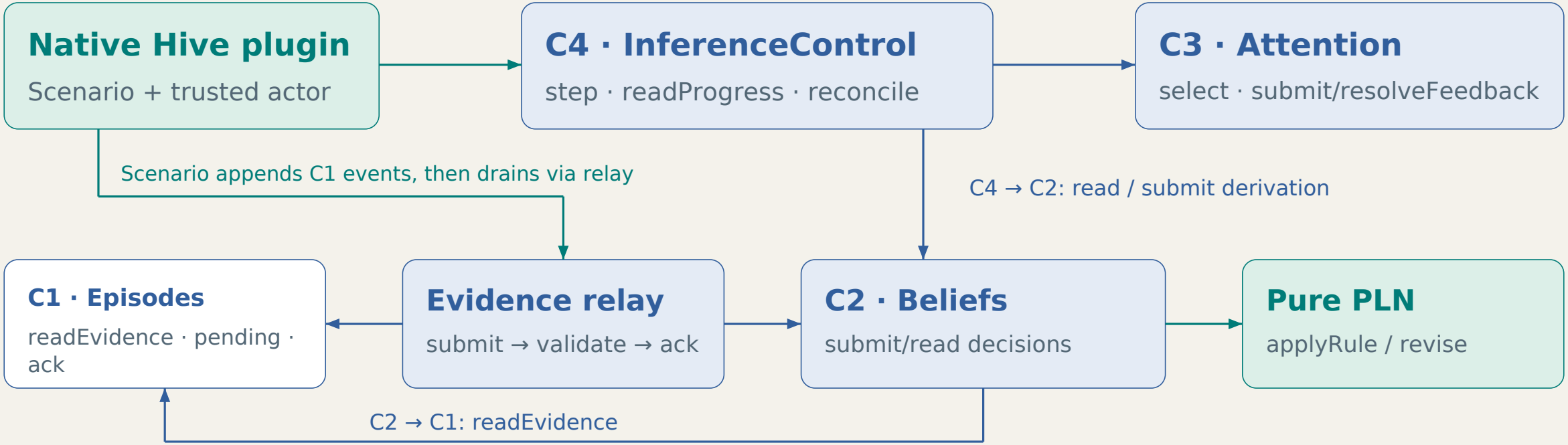
Control responsibilities: move work, preserve authority

The remaining six modules connect owners and report specific facts.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / STATUS
Messaging	Accept, deliver, acknowledge send · poll · ack	Inbox + deduplication · IMPLEMENTED
Execution	Run one named operation submit · status · result	Jobs + completion outbox · IMPLEMENTED
Registry	Describe configured providers descriptors	Static configuration · IMPLEMENTED
Resolver	Match required semantics resolve	No mutable state · IMPLEMENTED
Consensus / finality	Submit approved commitment; observe submit · observe	Signed intents + observations · REAL M4
Authorization	Check current caller + scope + method authorize / live policy checks	Operator policy · LOCAL FIXTURE

Function dependencies: the exercised M3.1 path

Arrows point from caller to provider; the relay and every state owner commit separately.



C1 / C2 / C4 → private AtomStorage → real DeTTa

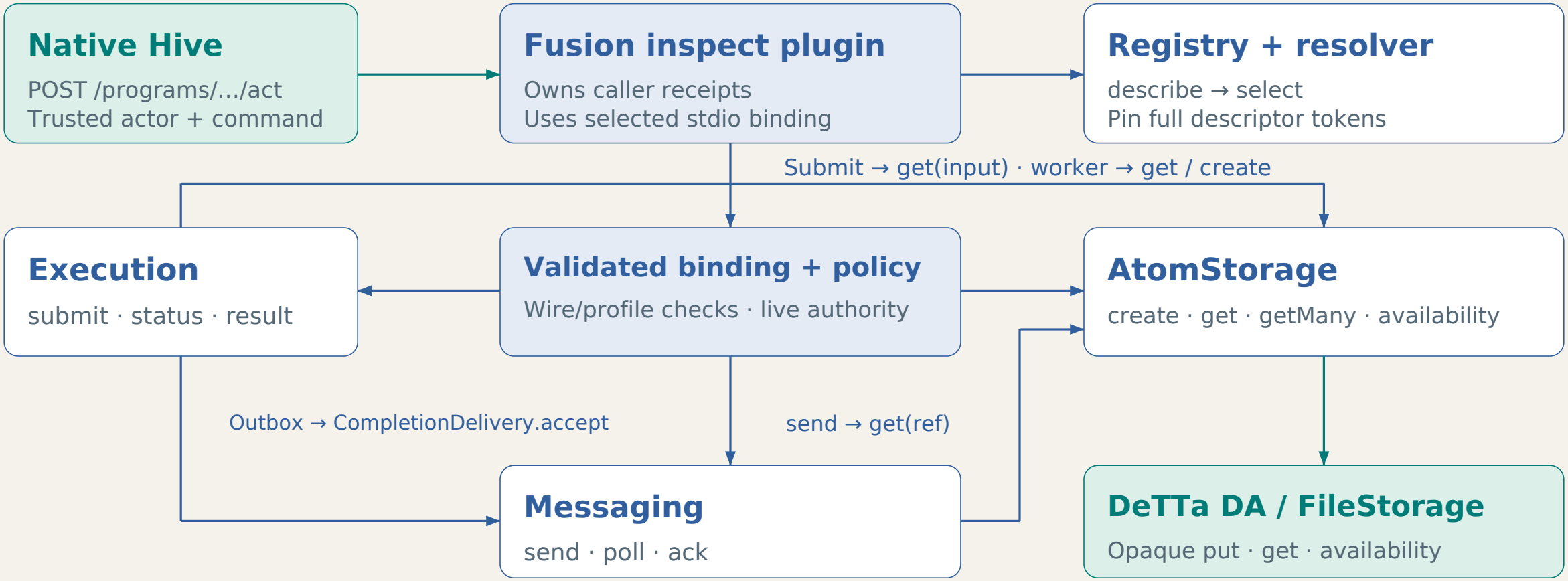
Opaque bytes below the application boundary; each owner keeps its own journal.

Authority + descriptors

Checked at every owner port

Function dependencies: the preserved M1.2 path

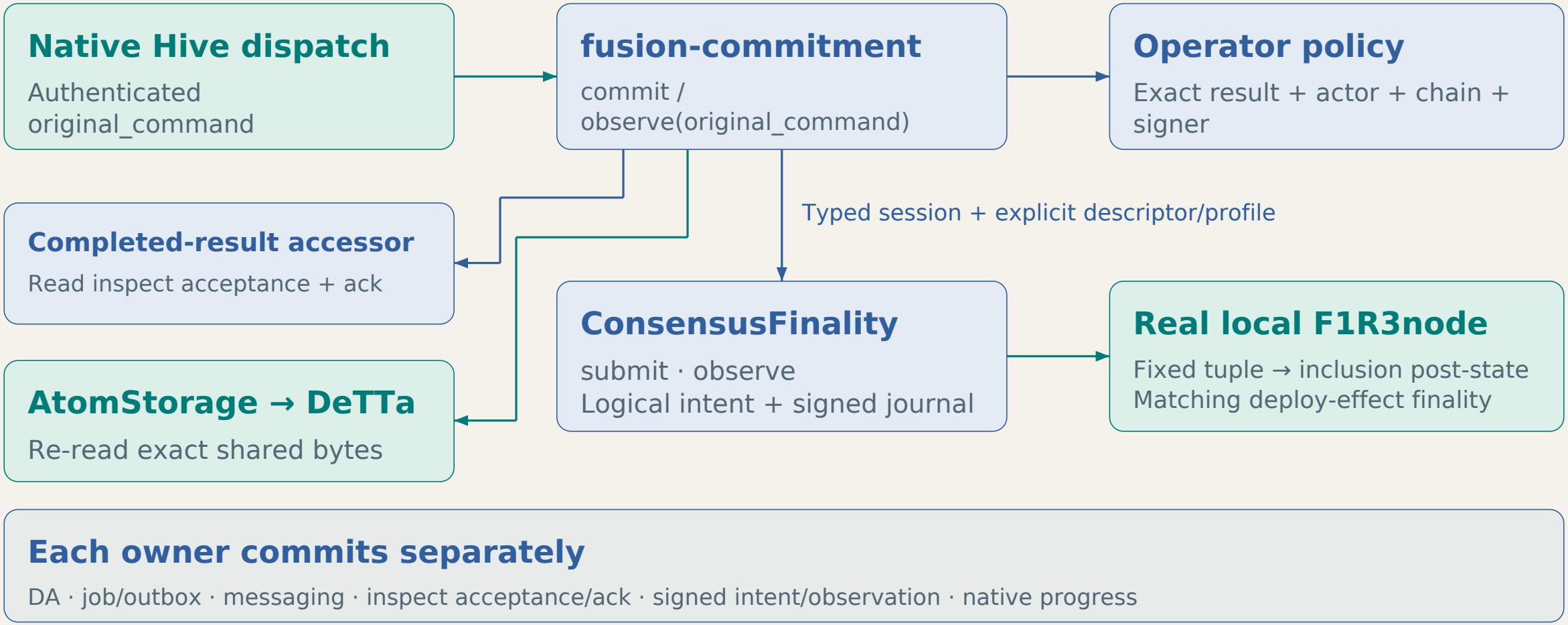
Solid arrows = exercised calls. Every provider effect crosses an existing validated interface.



M4 adds a separate commitment caller after acceptance; this original inspect path stays unchanged.

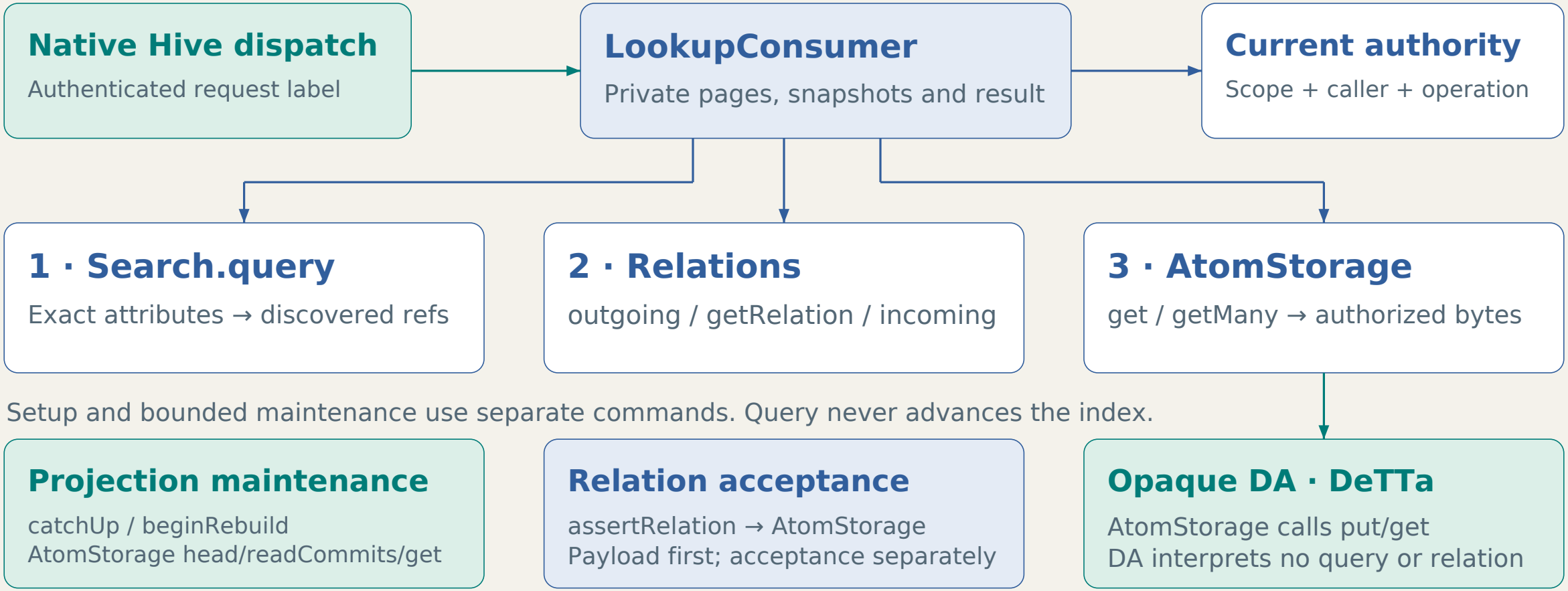
Function dependencies: an approved result reaches finality

A native sibling consumes the completed inspect result; private cognition has no chain dependency.



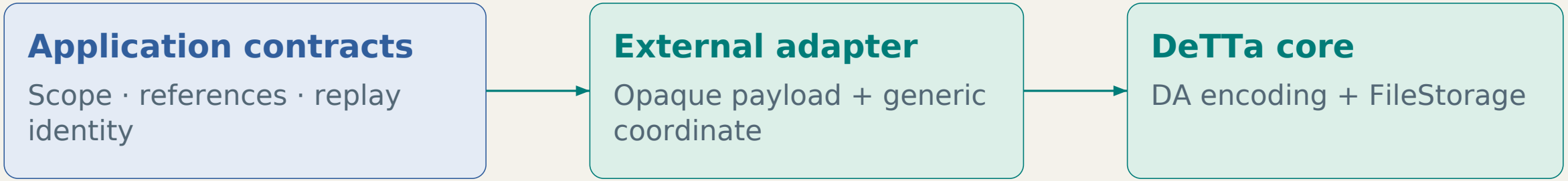
Function dependencies: discover → traverse → retrieve

Solid arrows show caller-to-provider calls; each owner commits its own state.

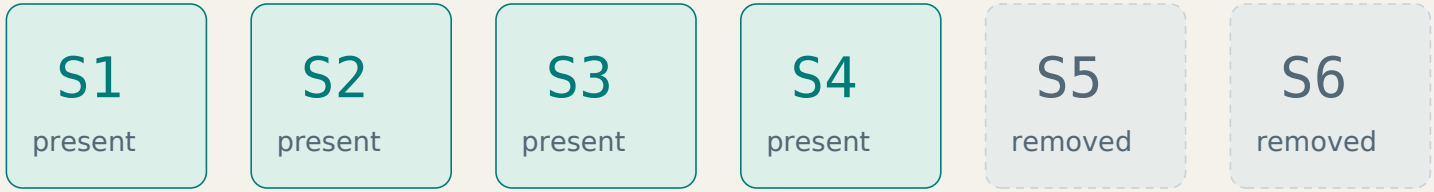


The DA boundary remains application-neutral

The external adapter translates application references into generic coordinates and opaque bytes.



4 data shares + 2 parity shares



Observed reconstruction

Cached payload + two shares removed: exact bytes recovered. A third missing share is unavailable.

The DA core does not interpret Hive identities, truth values, attention balances or job state.

Independent replacements preserve the caller contract

Select explicit profiles; rebuild derived state through the authoritative source.

AtomStorage · M1 / M5

Real DeTTa-backed owner or atomic-file fixture. The M5 feed remains ordered and replayable.

Attention · M3

SQLite or atomic-file fixture. Same STI policy, tasks and budgets; same consumer outcomes.

Search · M5

SQLite or atomic-file fixture. Same corpus and prefix; independent predicate evaluation and persistence.

The Search fixture still answers when the primary predicate helper is disabled.

Fresh stores and explicit profiles; accepted state is not migrated.

Private state has explicit owners and placement

A trusted Hive actor maps to an operator-configured private scope and credential.

PRIVATE LOCAL COMPOSITION

C1 · Episodes

History + delivery intents

C2 · Beliefs

Supports + exact attempts

C3 · Attention

STI + clock + causal replay

C4 · Controller

Budgets + progress + feedback

Public Hive receipt

Generated counters, status and public command receipt ID.

Private trace

Task/support hashes, premises, lineage, attention and payloads stay private.

Shared M1 wire and DA contracts stay unchanged; every owner rechecks current authorization.

Executable contracts make the inference step finite

The supported task is explicit, immutable and grounded before the controller starts.

CONTRACT	WHAT IS FIXED	WHY IT MATTERS
Context + task	Private context; exact role references	Labels cannot merge distinct identities
Workset	1–16 unique immutable tasks	No search or automatic task expansion
Premise snapshot	Claim / context / revision / support ID	No silent substitution with latest state
Derived support	Closed tag + inherited roots / ancestry	No fresh observation or evidence count
Work budget	Candidates, attempts, generation, deadline	Restart cannot silently refill the allowance

Profile mismatch, malformed response or changed semantic intent fails explicitly.

C1: one occurrence, one position in the story

Immutable event atoms sit behind a stable episode identity and a committed sequence.

EpisodeRef = scope + ID

1 · Report

No numerical claim

2 · Assessment

Explicit strength / confidence

3 · Copy

Inherits original roots

4 · Contrary

Kept in the history

Same occurrence + same body

Return the original receipt, even under a new request key. The head does not advance.

Same occurrence + changed meaning

Conflict on body or provenance changes. Distinct real occurrences may share identical bytes.

C1 commits event + evidence intent together; AtomStorage has already committed independently.

C1 delivery closes without understanding PLN

Completion attests durable processing. Belief support and numerical truth remain C2 responsibilities.

C1 event
Event + delivery intent

→

C2 decision
Full selected-profile result

→

Relay decision
Validate + save completion

→

C1 acknowledgment
Exact identity + live consumer

→

Relay acknowledgment
Save progress; recover replies

33 events · one persistent episode

Append 16 → drain to 0
Append 16 → drain to 0
Append 1 → drain to 0

Acknowledgment is model-independent

The trusted relay checks C2's full decision. C1 checks its event, delivery, request and selected consumer; it parses no PLN snapshot.

16 pending deliveries is still the bound. Draining frees pending capacity; episode lifetime stays 256 events.

C2: admitting evidence and changing belief are separate

New observations remain visible even when they cannot change the current aggregate.

Evidence submission

C1 evidence reference or an existing C2 support; exact claim and context.

Admission decision

ADMITTED or DUPLICATE
Track occurrence and provenance.

Numerical decision

Initialize, revise, no change, retain alternative, or model-inapplicable.

Aggregate with exact coverage

Every numerical result names the contributing leaves, origins, premises, formula and assumption.

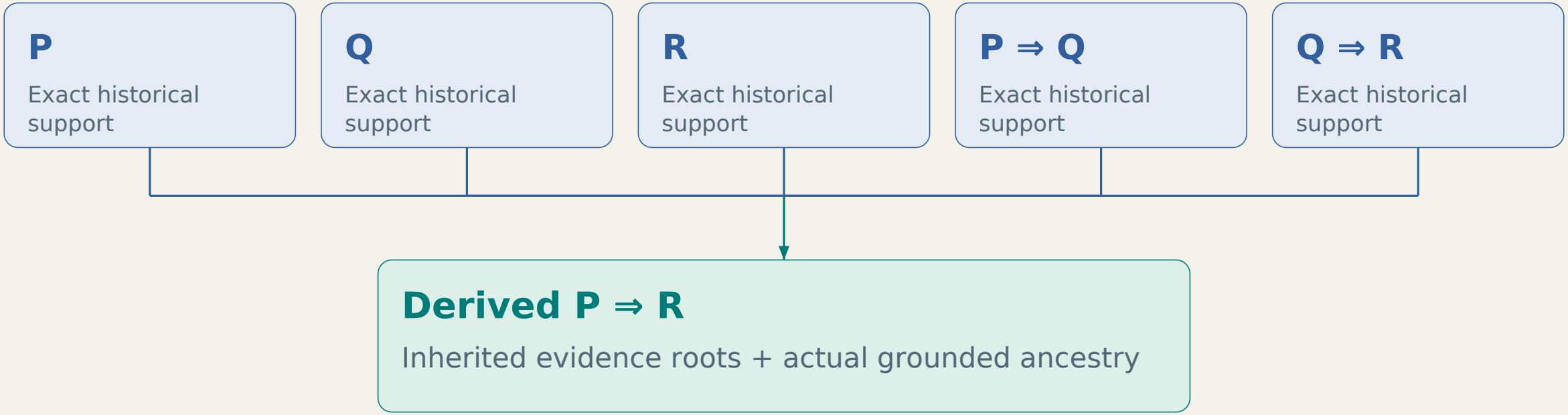
Unaggregated alternatives

Copied or contrary observations remain in the snapshot. Later eligible assumptions can combine existing supports.

One saved truth value is not a summary of every observation. Read its coverage and alternatives.

A derived conclusion inherits its evidence trail

The result is a new support record, while its original observations and ancestry remain the same.



Unknown target: initialize inferential support. Existing truth: keep the alternative.

No new observation, root, independence assertion or evidence count is created.

One grounded rule reuses the actual native PLN

No replacement arithmetic and no general search loop: five explicit premises produce one proposal.

Fusion PLNKernel

Exact bindings, source/profile checks and remaining deadline

→

Pinned Reachability

PLNAdapter.apply_rule
DeductionRule + guards

→

Native PLN / PeTTa

Truth_Deduction
SWI 10.1.14 + Janus

Two probability checks

Upstream exact Python precheck and the native consistency guard both apply.

Separate compatibility profile

Deduction has its own source/runtime allowlist; the revision-only profile stays intact.

Kernel output is a proposal. C2 performs final authorization, CAS and admission.

Native goldens check deduction and preserved revision

Independent expected values check pinned native formulas; no replacement math runs in the owners.

M3 · Grounded deduction

P, Q, R = (0.5, 0.8)
P⇒Q, Q⇒R = (0.8, 0.7)

Expected P⇒R: (0.68, 0.3136)

M2 · Guarded revision

(0.75, 0.50) + (0.25, 0.50)
→ (0.50, 2/3)

Unequal confidence 0.50 / 0.75
→ (0.375, 0.80)

`Observed deduction: 0.6800000000000002 / 0.31360000000000005`

Absolute tolerance 1e−12; finite-domain and lineage guards remain active.

These are model-specific heuristics, not calibrated real-world confidence or proven entailment.

C3: a small, independent attention policy

sti-additive-linear-decay/1 · integer STI · explicit logical time · persistent causal identity

Stimulate

Add within per-target saturation. Refuse the whole change if the context budget would be exceeded.

Select

Rank positive eligible targets by descending STI, then the full canonical reference. Reads do not decay state.

Advance

Explicitly decay by logical steps. Stored causal generation survives decay, new ingress and new step IDs.

Default limits: 100 STI per target · 300 per context · decay 1 per logical step

Only authority and opaque-reference syntax are required; no Hive, storage or PLN runtime.

This reference policy is implemented. ECAN, LTI, diffusion and new attention research are out of scope.

Attention replacement preserves the measured outcome

Recorded M3 comparison: same tasks, evidence, policy and budgets; provider and preference vary.

ATTENTION PROVIDER	PREFER	SELECTION ORDER	STEP TIME
SQLite	B	b → a	0.744361 s
SQLite	A	a → b	0.677157 s
Atomic-file fixture	B	b → a	0.726126 s
Atomic-file fixture	A	a → b	0.641150 s

Every case

1 native call · 1 unique admission · 1 feedback credit

Reopen all owners

Same receipt · zero additional native calls

Four local observations demonstrate substitution and selection effect; they are not a performance ranking.

C4: reserve work durably before invoking the rule

A work budget limits physical evaluation opportunities, even when the process dies.

1 · Freeze selection

Read the immutable workset and exact attention order.

2 · Reserve attempt

Commit candidate progress and a physical attempt slot.

3 · Ask C2

Recover a decision or dispatch the exact derivation attempt.

4 · Reconcile

Save decision and causal feedback acknowledgment.

Budget survives restart

Candidate count, reserved attempts, generation limit and absolute deadline are persisted.

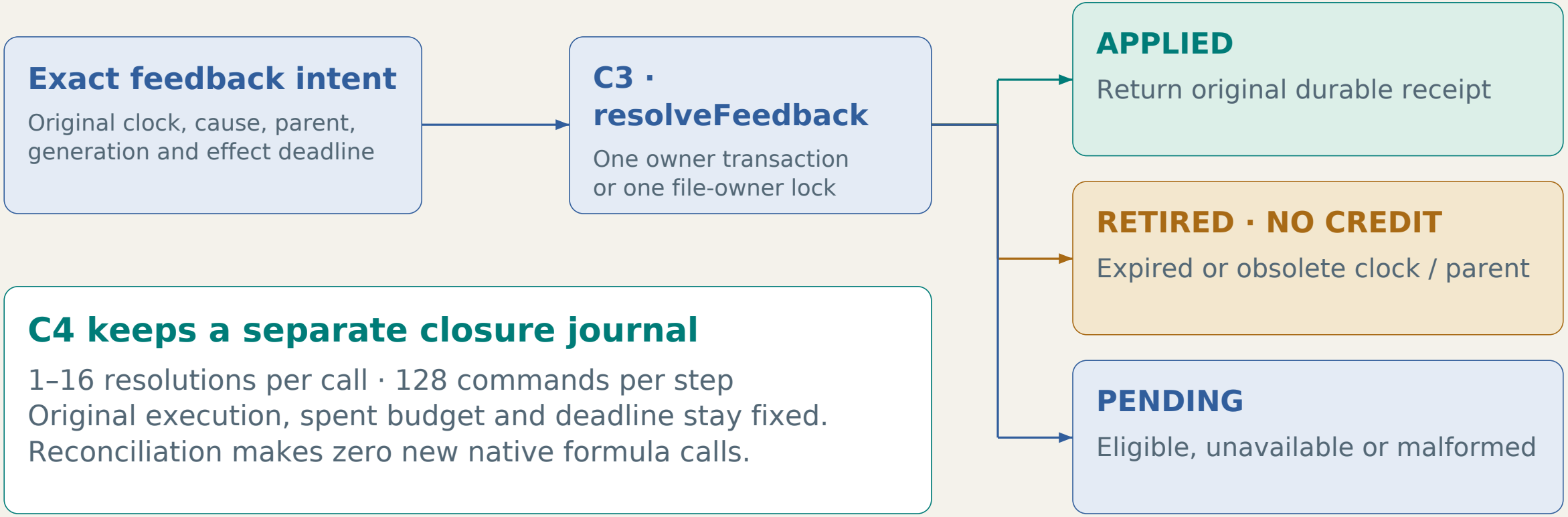
Selection changes next step

Current selection is frozen. A unique admission records one causal feedback intent.

No refund for a lost pending attempt. A new evaluation needs another slot and attempt identity.

C3 alone settles the feedback outcome

C4 asks about the exact recorded intent. Attention serializes application and retirement.



Applied before a lost reply → recover once. Retired first → delayed delivery cannot award credit.

Five crash boundaries, independent owner commits

10 direct boundary cases + 5 actual Hive caller boundary cases in the recorded M3 run.

PROCESS DIES AFTER…	DURABLE FACT	RECOVERY RESULT
Attention stimulus commit	C3 causal receipt	Replay does not add STI again
C4 attempt reservation	Slot already spent	One-slot budget can end without a result
Kernel return; before C2 commit	Pending attempt; no admission	Another evaluation needs another slot
C2 belief commit	Authoritative saved decision	Recover without another native call
C3 feedback commit; before C4 ack	Credit recorded by logical cause	Recover ack without a second credit

The evidence proves tested local process recovery, not a distributed transaction or power-loss durability.

Authorization, replay and profiles remain live

Cached success does not bypass current authority, and compatibility is never silently widened.

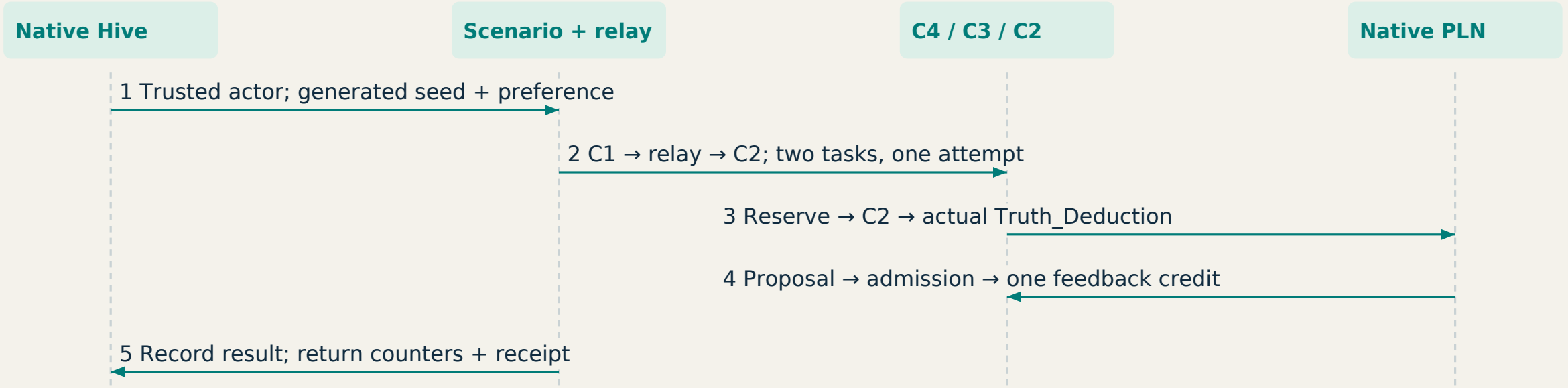
TRIGGER	OBSERVED DECISION	PRESERVED BOUNDARY
Same key, altered nested request	CONFLICT / validation refusal	Exact immutable semantic intent
Permission revoked before replay	DENIED	Live authority even for cached receipts
Authority / CAS changes during math	Refuse final admission	C2 rechecks before its own commit
Runtime or response binding changes	Explicit compatibility failure	No weaker fallback or invented result
Clock advances before feedback	Retire without credit via owner	Original clock, parent and generation retained

Attention ingress and feedback have distinct trusted bindings; selection never grants authority.

One real Hive action reaches the entire private path

Existing authenticated Programs runtime → fusion-inference / infer-generated.

```
POST /api/programs/fusion-inference/act
```



One private episode: 10 generated observations + action + result. Public receipt contains no private trace.

Native Hive exposes execution and closure separately

infer-generated → restart → reconcile-generated(original_command, limit)

Original execution · immutable

One examined candidate
One reserved / completed attempt
One unique admission · zero feedback applied

Later closure · versioned

Same original result and counters
Evidence backlog and feedback outcomes
Stale feedback retired without credit

Lost reply after feedback applied

Recover the original receipt after expiry or decay, even when current STI has reached zero.

Expired before execution

Hive can finish the action with zero admissions and zero reservations. Those counts stay explicit.

Public done / committed means the action returned. Detail reports execution, evidence and feedback closure.

The same one-attempt budget makes attention visible

Two eligible tasks have exact premises. Initial STI selects which one can consume the attempt.

Prefer task A

A	20
B	10

First attempt → A

Prefer task B

A	10
B	20

First attempt → B

Both attention providers preserve this effect; replay adds no native call and no second credit.

Attention changes scheduling. It does not alter premise truth values or manufacture independent evidence.

One artifact connects Hive, DA and the chain

The approved result is the exact result retrieved, committed and independently read back.

53-byte input

Generated and explicitly shareable
Real native fusion-inspect

→

143-byte result

Real worker + real DeTTa
Caller accepts and acknowledges

→

One commitment

Approved digest in real post-state
Matching finalized deploy effect

APPROVED RESULT SHA-256 · SAME VALUE AT EVERY STAGE

`cd2f727689b41e243ee78207d058d00b8be64932abbfd86a02851dfcbac400bb`

Counts remain stable

1 job · 1 outbox · 2 messages · 2 shared atoms

First finalized-effect observation

4.066 s · one native observe call · 675 phlo

These are local observations from one run, including proposal/observation overhead; no throughput claim.

Approval freezes the result and its destination

The public action accepts only original_command. Trusted composition supplies all sensitive choices.

Exact result

Generated command + seed + shared scope
Accepted result reference and digest

Exact authority

Authenticated actor + swarm + expiry
Live checks on new calls, replay and send

Exact destination

Source/profile + network + shard + genesis
Registered signer and protobuf identity

Exact operation

Fixed tuple and gas limits
Canonical copies detach mutable inputs

No caller-supplied code, digest, endpoint, signer or approval; revocation also blocks cached replies.

Finality is tied to the actual stored effect

Submission, inclusion and verified finalized effect are different observations.

1 · Submission

The node accepts the saved signed deploy. The effect is still pending.

2 · Inclusion

The exact deploy appears in a block with successful execution and cost.

3 · Verified effect

Read the exact tuple in that block’s post-state; match the deploy’s finalized effect.

Fixed on-chain value

(namespace, generated operation ID, approved result digest)

Independent verification

Same digest + deploy + block + post-state after node restart; no caller journal or signer.

The adapter verifies node-observed evidence. The chain does not certify off-chain PLN reasoning.

A lost reply reuses the same signed request

Persist the logical plan before signing, then persist the complete signed deploy before HTTP submission.

KILL BOUNDARY	AUTHORITATIVE DURABLE FACT	OBSERVED RECOVERY
After result acceptance	Caller acceptance; completion may need ack	Acknowledge same result, then approve
After signed-intent commit	Exact signed bytes and timestamp	Replay with signing key unavailable
After node send, before receipt	Original signed deploy identity	Retry after real finality; same request
After effect, before progress	Chain observation committed separately	Native caller recovers that observation

Node outage preserves the off-chain result. Restart the same genesis/data and read the same finalized effect.

The real chain ran in a disposable isolated network

Clean pinned F1R3node source; no public connection, upstream patch or consensus rewrite.

UNPRIVILEGED NETWORK NAMESPACE · LOOPBACK ONLY

Generated test genesis

One validator
Generated stake, wallet and key

→

Real F1R3node

9ec1096e1077…
Fresh source and local build

Observed sockets stay inside

P2P and gRPC wildcard listeners exist only in the namespace; no route to an external network.

From the host

All six probed ports refused connections while the node was ready.

From the namespace

External connection failed: ENETUNREACH. Host firewall unchanged.

Single-validator evidence establishes this local integration; multi-validator safety remains unrun.

Four Relations operations expose accepted structure

Relation meaning includes schema, direction, ordered roles and private scope.

OPERATION	RESPONSIBILITY	OBSERVED SEMANTICS
assertRelation	Persist payload, then accept structure	Semantic duplicates converge; changed replay intent conflicts
getRelation	Read an accepted relation	A payload orphan alone is never an accepted relation
outgoing	Page source-role incidence	Fixed accepted head; query-bound continuation
incoming	Page target-role incidence	Reverse incidence verified by the native consumer

requires: dependent → prerequisite · transfer: 3 roles · association: symmetric

Role order is preserved. Repeated endpoints and self-links are explicit, with each incidence returned once.

A committed source makes the index rebuildable

The source enumerates accepted immutable envelopes; Search interprets one declared content type.

AtomStorage owner

Accept immutable envelope + feed metadata in one local commit. Source UUID, scope and contiguous sequence identify the prefix.

Bounded maintenance

Read head/readCommits, retrieve bytes with get, validate each envelope, then extract exact attributes. At most eight new records per call.

Search owner

Commit derived index rows + checkpoint together. Replay resumes from that committed prefix. beginRebuild creates a new generation.

Fail without advancing

Gaps, unavailable bytes, forged metadata or changed anchors block the batch; no invented prefix.

No foreign owner database is read. The alternate provider consumes the same declared source.

A complete page is not a promise of current knowledge

Search returns separate, inspectable claims about what it has actually indexed.

CLAIM	MEANING	WHY IT IS SEPARATE
Indexed prefix W	Every accepted source item through W processed	Later source commits may exist
Complete page	All matches for this page at its fixed prefix	A continuation can still be required
Requested minimum	W meets the caller’s source barrier	Too far behind → NOT_AVAILABLE
Observed source head H	Whether W equals H, with observation context	No query-side head poll or ingestion

Exact conjunction: 1–3 distinct fields from kind / topic / label; equality only.

Case and Unicode are exact; missing fields do not match. Search W and Relations R are separate snapshots.

Actual Hive discovers the answers through the interfaces

Setup exits before lookup; the consumer receives query attributes and a freshness barrier only.

5
matching roots

→

15
accepted relations

→

8
unique payloads

TRANSITION	NATIVE BEHAVIOR	EVIDENCE
Initial query	3 pages at page size 2	Distractors and another private scope excluded
Append → lag → catch up	NOT_AVAILABLE → 6 roots / 9 payloads	Query did not ingest; maintenance advanced prefix
Rebuild → restart → substitute	Same discovery and canonical bytes	New generation; independent file Search fixture

The oracle is test-only and removed before startup. No answer references are supplied to Hive.

Thirteen process kills exercise separate commit boundaries

A lost response does not merge owner transactions or erase accepted work.

BOUNDARY	DELIBERATE SIGKILL CASES	RECOVERED FACT
Atom / feed acceptance	1	Accepted envelope and contiguous feed stay together
Relation payload / acceptance	2	Orphan is invisible; accepted relation replays once
Projection checkpoint / rebuild	6 · both Search providers	Before/after checkpoint and rebuild intent resume
Native page / result progress	4 · both Search providers	Page continuation and completed result survive

Same profile, independent Search implementation, unchanged consumer; fresh derived store for substitution.

Private ownership extends without changing cognition

Each new profile is explicit; old M1–M4 owners and behavior remain selectable.

OWNER	STATE AND COMPATIBILITY	PRIVATE BOUNDARY
AtomStorage	immutable-commit-feed/1 · contract 1.1.0	Envelopes + accepted source metadata
Relations	schema-ordered-relations/2 · contract 2.0.0	Accepted structures + replay + incidence
Search	exact-attributes/1 · contract 2.0.0	Derived index + checkpoint + generation
Native lookup caller	Request + pages + full authorized result	Operator request label; ordinary receipt = counts/status

Lookup has no runtime dependency on C1/C2, attention, inference or chain; no private identifiers are exported.

Coverage has three denominators

Module representation, exercised operations and behavioral coverage answer different questions.

15 / 15

Module families

Up from 13. All bounded reference families now have a represented implementation.

9 / 9

New operations

4 Relations + query + 2 feed + 2 maintenance. With inherited storage: 13 / 13 in this slice.

18 / 18

Behavioral groups

12 required scenario groups + 6 crash-boundary groups. Separate from test-method counts.

This is bounded reference coverage, not full Omega completeness. Line/branch coverage: NOT_MEASURED.

Real providers, fixtures and unrun work stay distinct

All required M5 gates pass; each claim retains its actual provider and source label.

CLASSIFICATION	M5 EVIDENCE	MEANING
REAL PROVIDERS	Authenticated native Hive / PeTTa · DeTTa SQLite AtomStorage, Relations and Search	Actual dispatch, persistence, discovery and restart
FIXTURES	Generated corpus / oracle / local policy Independent atomic-file providers; injected faults	Controlled inputs and substitutions; not production data or services
BLOCKED	No required M5 prerequisite Historical dirty M1 chain source unavailable	Historical blocker remains separate from the clean M4 rebaseline
UNRUN / OUT OF SCOPE	Power loss, distributed failover, empty caches Live migration, public networks, new chain work	No claims for these scenarios; code coverage not measured

The six real M4 chain cases passed again as legacy regression; M5 adds no chain operation.

Executed tests cover discovery, recovery and regressions

Repeated subsets overlap. These counts must not be added into a unique test total.

LAYER	CASES / GROUPS	RECORDED OUTCOME
New contract / fixture methods	18	PASS · pagination, freshness, auth, independent providers
Real DeTTa + owners	4 methods	PASS · includes nine direct SIGKILL cases
Actual authenticated Hive	7 cases	PASS · includes four native SIGKILL cases
Supplemental private probes	4 + 4	No cognitive effects; complete payload comparisons
M1–M3.1 regression	156 aggregate	PASS · includes old 138 plus new 18 methods
Unchanged M4 suites	11 fixtures + 6 real	PASS · real isolated chain and independent readback

Tested bfb0ec8 · 13 deliberate M5 SIGKILL cases · zero final failures, blockers or timeouts.

M4 gates: native result to verified chain effect

All 10 required gates passed. CHAIN is required and real in this milestone.

GATE	RESULT	RECORDED EVIDENCE
BASELINE	PASS	Full M1–M3.1 regression; both attention providers and old profiles
CHAIN_SOURCE_BUILD	PASS	Clean pinned build; loopback-only namespace; live socket probes
NATIVE_HIVE_DA	PASS	Actual inspect dispatch + real DeTTa + labelled fixture counterparty
APPROVAL_BINDING	PASS	Exact immutable result/chain/profile; current authority and refusal cases
REAL_CHAIN_EFFECT	PASS	Actual inclusion post-state tuple + matching finalized deploy effect

No missing prerequisite, final executed failure or positive observation timeout is hidden in PASS.

M4 gates: recovery, privacy and independent readback

All 10 required gates passed. CHAIN is required and real in this milestone.

GATE	RESULT	RECORDED EVIDENCE
RECOVERY	PASS	Four caller/adapter kill boundaries; real node outage and restart
PRIVACY_ISOLATION	PASS	Private canaries and identifying derivatives absent; live isolation
REPRO_READBACK	PASS	Fresh pinned builds + independent reader checking the same artifact
DEMO_HANDOFF	PASS	Native receipts/progress, runnable commands and sanitized trace
SCOPE	PASS	Original DA, cognition, math, ports and inspect plugin preserved

No missing prerequisite, final executed failure or positive observation timeout is hidden in PASS.

M5 gates: contracts, real storage and rebuild

All 12 required gates pass on the exact final tested implementation.

GATE	RESULT	RECORDED EVIDENCE
BASELINE_AND_SCOPE	PASS	Requested ancestry; clean source; preserved M1–M4 boundaries
RELATIONS_CONTRACT	PASS	All four operations; ordered roles, schemas and fixed-head pages
RELATIONS_REAL	PASS	Accepted relations and payloads through actual local DeTTa
COMMITTED_SOURCE	PASS	Replayable, scoped AtomStorage-owned contiguous feed
SEARCH_CONTRACT	PASS	Exact attributes; explicit freshness, pagination and error semantics
SEARCH_REAL_REBUILD	PASS	Bounded catch-up and generation rebuild from real source

Final missing prerequisites: 0 · executed failures: 0 · observation timeouts: 0.

M5 gates: substitution, recovery and actual Hive

All 12 required gates pass on the exact final tested implementation.

GATE	RESULT	RECORDED EVIDENCE
SUBSTITUTION	PASS	Independent feed/Search fixtures; unchanged declared-port consumer
RECOVERY	PASS	13 deliberate process kills, replay and durable owner boundaries
AUTH_PRIVACY	PASS	Live authorization, private placement and refusal coverage
HIVE_LOOKUP	PASS	Actual authenticated native query → traversal → payload retrieval
REPRO_AND_LEDGER	PASS	Pinned recipes; separate module/operation/behavior coverage
LEGACY_REGRESSION	PASS	M1–M3.1 and unchanged M4 fixture + real-chain suites pass

Final missing prerequisites: 0 · executed failures: 0 · observation timeouts: 0.

Source, build and outcome remain independently traceable

The final tested implementation and evidence-only descendant are explicit.

LAYER	RECORDED STATE	BOUNDARY
Requested / actual start	05b1c52 → 4093301	Presentation-only descendant before M5
Clean tested implementation	bfb0ec8856c9	Final native plugin and Python loaded from this source
Evidence snapshot	3dc6c1146601	Docs/evidence only; runtime and tests unchanged
Fresh native provider build	Hive 34b0fb3 · DeTTa 272888f	Fresh sources, venv and native objects; warm caches
Preserved math / chain	PLN 4405956 · F1R3node 9ec1096	Same formulas; reused pinned node binary

M5 archives final passes, corrected development failures and intentional negative classification separately.

The limitations are part of the contract

M5 adds bounded private discovery while preserving the prior cognition and chain limits.

Exact attributes; one-hop lookup

Case-sensitive conjunction over kind, topic and label. No embeddings, semantic search or task generation.

Fresh profiles and intact files

Finite local stores. No migration, eviction or automatic repair; unavailable source bytes block indexing.

Independent snapshots

Search prefix W and Relations head R are explicit. They do not form a global atomic snapshot.

Earlier guarantees retain their limits

Cognitive budgets stay spent; closure can remain pending. Chain proof remains one local validator.

Stopped at M5: no new cognition, embeddings, scheduler, migration or chain operation.

Discover the knowledge. Preserve the boundaries. Check the evidence.

Real Hive → exact query → relations → real DeTTa payloads
with explicit freshness, independent owners and restart.

M5 COMPLETE · ALL 12 REQUIRED GATES PASS

Implementation map: responsibilities stay local

Follow interfaces and owner journals before following concrete provider classes.

RESPONSIBILITY	FILES / SURFACE	OWNED STATE
C1 / relay / original C2	episodes.py · evidence_relay.py · delivery_contracts.py	History; delivery + trusted completion
C2 derived admission	derived_beliefs.py · cognition_contracts.py	Fresh derived-beliefs.sqlite · local v3
C3 attention	attention.py · adapters/fixture_attention.py	SQLite or independent atomic-file state
Pure PLN providers	adapters/pln_deduction.py · pln_revision.py	No mutable domain state
C4 controller	inference.py · deadlines.py	Budgets / execution + separate closure journal
Composition / native caller	cognition.py · hive/inference_program/	Explicit provider bindings; caller receipt

Exact sources identify the verified native stack

Full revisions are printed here; source files and built binary hashes remain in the committed evidence.

Fusion · tested	`bfb0ec8856c932dc2bcbc05b8949c957da8d72cd`
Hive schematics	`4af00db4e395375484dd48d0b9068a59afcf01f8`
Hive	`34b0fb32ab3806e5609fc89bb7535518a35e1aea`
DeTTa	`272888f264dd2dde18b574cabb53c266b0dac2ec`
Reachability	`dd56f3f053ed90239de5377021a7d3b3fbf424ac`
PLN	`4405956947c4b53c7ff01bd565aa3b114bc970a1`
PeTTa	`ae66fa8e41dcd5539d614706bd4e5cfb34f9608d`
SWI 10.1.14	`69775434c8226897626b226aefcc8266499f1e2e`
F1R3node	`9ec1096e10774926ea8e72f0823482f0bf9213e1`

metta-attention inspected at 3e1c8147; not used as the bounded STI provider.

Profiles and bounds make compatibility explicit

fusion-delivery-closure/1 selects fresh C1 / C3 / C4 journals; C2 and the math stay unchanged.

PORT / PROFILE	EXPLICIT SELECTION	KEY BOUND
C1 · local 3.0.0	episode-delivery-completion/1	16 pending; 256 events; trusted completion only
C2 derived · local 3.0.0	pln-derived-private/1 · unchanged	Historical premises and inherited support roots
C3 · local 2.1.0	sti-additive-linear-decay-closure/1	Existing STI, cause and command capacities
C4 · local 2.1.0	bounded-grounded-closure/1	1–16 resolutions/call; 128 commands/step
PLN kernel · local 2.0.0	grounded-five-premise-deduction/1	One unchanged native rule; five exact premises
Historical M1 / M2 / M3	Original descriptors remain selectable	No silent upgrade; no live state migration

New profiles close the seam; old stores stay explicit

The M3 findings were reproduced before edits. M3.1 is an opt-in contract extension.

Historical M3 profile

C1 expects a closed v2 acknowledgment.
C2 emits a v3 decision.
Delivery remains pending; stale feedback can remain CONFLICT.

Fresh M3.1 profile

Relay validates C2; C1 gets narrow completion.
Attention resolves original feedback.
C4 tracks closure separately from execution.

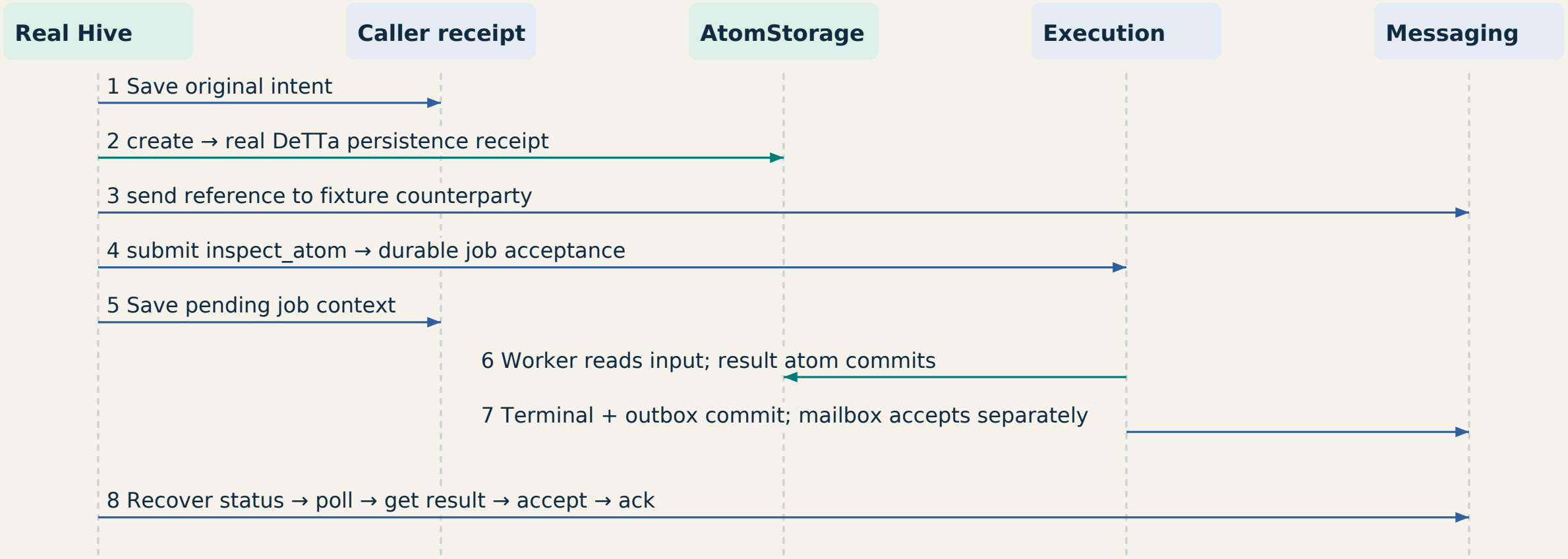
Populated old stores → new selection: UNSUPPORTED_PROFILE

Disposable copies reject without modification and still replay under the original profile. User stores are untouched.

No automatic conversion, repair of pre-existing pending work, invented acknowledgment or weakened validation.

One native operation crosses the existing interfaces

The caller stores intent; each provider commits only its own state.



Historical M1: one result reached an isolated chain

This is the earlier protocol-peer artifact. No M1.2, M2, M3 or M3.1 Hive result was submitted.

PENDING
Signed deploy accepted

→

INCLUDED
Block observed; effects pending

→

FINALIZED
Deploy effects reported finalized

M1 recorded observation

Block height	**3**
Execution error	**false**
Rejection count	**0**
Execution cost	**351 phlo**

Historical commitment digest

```
f9236451ff785dc58315251cf864a3d1
0938412551cf45f448055061bc953ea1
```

One local validator; direct node observations.
Not a consensus proof or production test.

Historical M1 reproduction stays BLOCKED. M4 separately passes with a verified clean source baseline.

Reproduce the required real M4 gates

Use tested 55ed540 or a reviewed descendant; fresh build/evidence destinations and authorized source access.

BUILD PINNED HIVE / DETTA / PLN + CLEAN F1R3NODE

```
uv run --python 3.11 --locked python \
scripts/build_m4.py <FRESH_BUILD>
```

VERIFY REGRESSIONS + NATIVE CHAIN + RECOVERY + READBACK

```
uv run --python 3.11 --locked python \
scripts/verify_m4.py <FRESH_BUILD> --output <FRESH_EVIDENCE>
```

Read: manifest → same-artifact → readback → verification → regression summaries

Separate reader: scripts/readback_m4.py · DA get-only authority + read RPC · no caller journal or signer.

Retry identities answer different questions

Preserve the logical cause while distinguishing transport replay from physical evaluation attempts.

IDENTITY / STATE	MEANING	RECOVERY ACTION
Request / step ID	Exact immutable command intent	Same body replays; changed body conflicts
Logical derivation ID	The same exact grounded conclusion	Deduplicate admission and causal feedback
Attempt key + reserved slot	One physical evaluation opportunity	Pending death spends slot; new key needs budget
Committed C2 decision	Saved authoritative admission outcome	Read first; do not recalculate
C3 causal feedback receipt	One workflow credit for logical admission	Recover acknowledgment; do not credit twice
Deadline / generation watermark	Durable limits across process restarts	Do not reset, refund or implicitly retime

M4 adds one profile and two narrow native actions

Existing ConsensusFinality signatures and legacy providers remain unchanged; fresh journals are required.

SURFACE / OWNER	CONTRACT OR CODE	DURABLE RESPONSIBILITY
Native commitment caller	integration/hive/commitment/	Original command mapping; public progress
Completed-result accessor	integration/hive/program/completed.py	Read caller-owned acceptance and ack
ConsensusFinality adapter	fusion/adapters/f1r3_m4.py	Plan → signed intent → receipt → observations
Signing boundary	fusion/adapters/f1r3_signing.py	Pinned serialization + registered local signer
Explicit compatibility	f1r3-generated-post-state/2	f1r3-local · interface 1.0.0 · impl 0.4.0
Bounded native observation	≤120 calls; stop starting after 180 s	≤256 saved observations; explicit exhaustion

Reproduce M5 from declared sources and private requests

Fresh provider build, existing chain dependencies, then all required gates.

```
uv run --python 3.11 --locked python scripts/build_m3.py <providers>
uv pip install --python <providers>/native/venv/bin/python \
-r integration/chain/requirements.lock
uv run --python 3.11 --locked python scripts/verify_m5.py \
--environment <providers>/environment.json --chain-build <chain> \
--output <fresh-output>
```

CODE	RESPONSIBILITY
fusion/atom_feed.py · relations.py · search.py	Accepted source; structural owner; projection owner
fusion/lookup.py · lookup_composition.py	Declared-port consumer; private binding and placement
integration/hive/lookup/ · test_m5_integration.py	Actual native dispatch and recovery; no answer refs

Use --gates hive for the focused demo. Keep raw environments, private journals, queries and IDs private.